

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
# Importing of necessary libraries
import numpy as np
import pandas as pd

import tensorflow as tf

from keras.models import Sequential

import matplotlib.pyplot as plt
import shutil
import os
import cv2
import PIL as IMAGE
```

```
# Creating the path for the archive file and the extracted file for unzip
archive_file="/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2.zip"
image_path="/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images"
```

```
# Unpacking the zip file with shutil library
shutil.unpack_archive(archive_file, image_path)
print(f"Successfully unpacked '{archive_file}' to '{image_path}'.")
```

1. Plotting some images from the loaded data

```
# Checking some images from each classes from the train dataset
import os
import random
import cv2
import matplotlib.pyplot as plt

# Path to dataset
train_path = "/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset"

# Number of samples per class
num_samples = 9

# Get class folders
classes = os.listdir(train_path)

for cls in classes:
    class_path = os.path.join(train_path, cls)

    if not os.path.isdir(class_path):
        continue

    images = os.listdir(class_path)

    # Randomly select images
    sample_images = random.sample(images, min(num_samples, len(images)))

    plt.figure(figsize=(4, 4))
    plt.suptitle(f"Class: {cls}", fontsize=14)

    for i, img_name in enumerate(sample_images):
        img_path = os.path.join(class_path, img_name)

        # Read image using OpenCV
        img = cv2.imread(img_path)

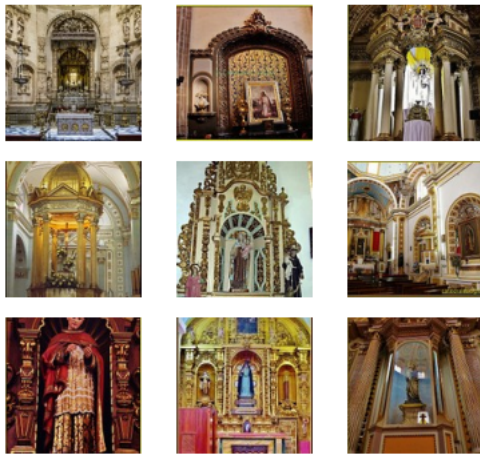
        if img is None:
            continue

        # Convert BGR → RGB (important for matplotlib)
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

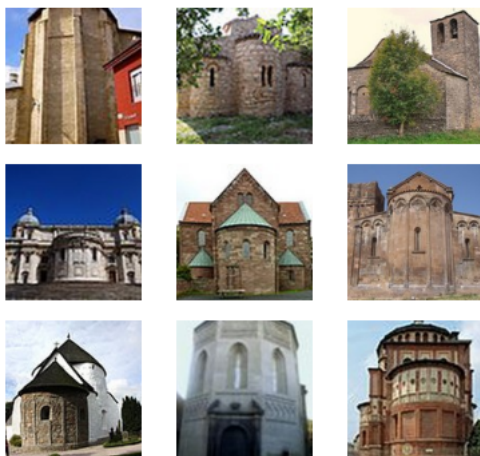
        plt.subplot(3, 3, i+1)
        plt.imshow(img)
        plt.axis('off')
```

```
plt.tight_layout()  
plt.show()
```

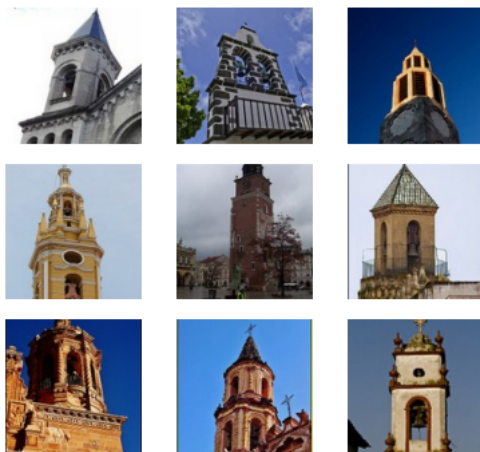

Class: altar



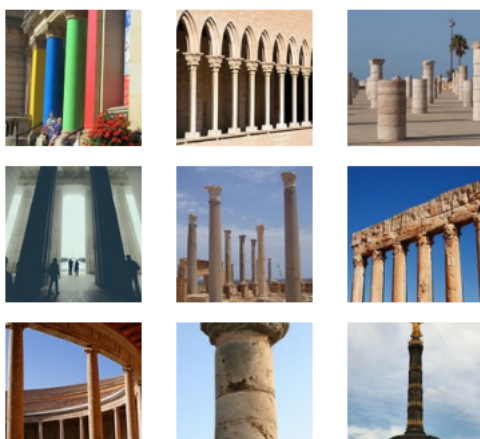
Class: apse



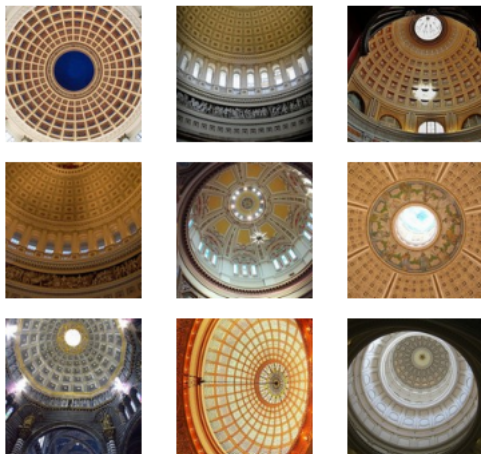
Class: bell_tower



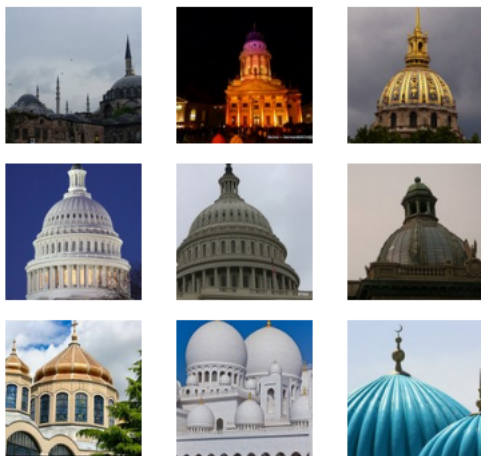
Class: column



Class: dome(inner)



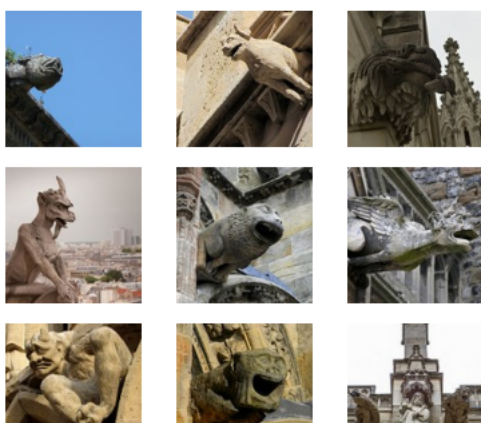
Class: dome(outer)



Class: flying_buttress



Class: gargoyle



2. Building the CNN model

Class: stained glass

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.applications import VGG16

print("TensorFlow version:", tf.__version__)
```

TensorFlow version: 2.19.0

```
# Some of the images may be distorted so we must remove those corrupted images
from PIL import Image
import os
test_path = "/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_
test_dir=test_path

def remove_corrupted_images(directory):
    bad_files = []

    for root, _, files in os.walk(directory):
        for file in files:
            path = os.path.join(root, file)

            try:
                img = Image.open(path)
                img.verify() # verify integrity
            except Exception as e:
                print(f"Removing corrupted file: {path}")
                bad_files.append(path)
                os.remove(path)

    print(f"\nTotal removed: {len(bad_files)} images")

remove_corrupted_images(test_dir)
```

Total removed: 0 images

```
# Image file generation without augmentation
# First defining the directory of train and test data set. Train data has been generated through the Image
# from the train data 20% of the data has been taken for the validation
# each class category has been encoded through the alphabetical encoding method

from tensorflow.keras.preprocessing.image import ImageDataGenerator

img_size = (128, 128)
batch_size = 32

train_dir = train_path
test_path = "/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_
test_dir=test_path
# Train + Validation generator
train_datagen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2
)

# Test generator (NO split, NO augmentation)
test_datagen = ImageDataGenerator(rescale=1./255)

# Train
train_gen = train_datagen.flow_from_directory(
    train_dir,
    target_size=img_size,
    batch_size=batch_size,
    class_mode='categorical',
    subset='training'
)

# Validation
val_gen = train_datagen.flow_from_directory(
    train_dir,
    target_size=img_size,
    batch_size=batch_size,
    class_mode='categorical',
    subset='validation'
)

# Test
test_gen = test_datagen.flow_from_directory(
```

```
test_dir,
target_size=img_size,
batch_size=batch_size,
class_mode='categorical',
shuffle=False
)

num_classes = train_gen.num_classes

print("Classes:", train_gen.class_indices)
```

```
Found 8193 images belonging to 10 classes.
Found 2042 images belonging to 10 classes.
Found 1475 images belonging to 10 classes.
Classes: {'altar': 0, 'apse': 1, 'bell_tower': 2, 'column': 3, 'dome(inner)': 4, 'dome(outer)': 5, 'flying_buttresc
```

```
# Removal of the corrupted file from the data
remove_corrupted_images(train_dir)
remove_corrupted_images(test_dir)
```

```
Removing corrupted file: /content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures
Total removed: 1 images
Total removed: 0 images
```

```
# print(train_gen.class_indices)
print("-----")
print(test_gen.class_indices)
```

```
{'altar': 0, 'apse': 1, 'bell_tower': 2, 'column': 3, 'dome(inner)': 4, 'dome(outer)': 5, 'flying_buttress': 6, 'tower': 7}
```

```
# Check some images and their class (encoded)
for i in range(5):
    print(test_gen.filepaths[i], test_gen.classes[i])
```

```

:one/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structures/Dataset_test/Dataset_test_original
:one/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structures/Dataset_test/Dataset_test_original
:one/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structures/Dataset_test/Dataset_test_original
:one/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structures/Dataset_test/Dataset_test_original
:one/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structures/Dataset_test/Dataset_test_original

```

```
print(test_gen.filepaths[200])    # full image paths
print(test_gen.classes[200])     # numeric labels
```

```
/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structure
```

```
print(train_gen.filepaths[:10])
print(train_gen.classes[:10])
```

```
['/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structu
[0 0 0 0 0 0 0 0 0 0]
```

```
train_gen.filepaths[900]
```

```
'/content/drive/MyDrive/Abhijit/Capstone/Capstone_Heritage/Images/dataset_hist_structures 2/dataset_hist_structu
```

```
# visualization one image and its label class from the train data
image_filepath = train_gen.filepaths[900]
label = train_gen.classes[900]
```

```
# Load the image using OpenCV
img = cv2.imread(image_filepath)
```

```
# Convert BGR to RGB (matplotlib expects RGB)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
plt.imshow(img)
plt.title(f"Label: {label}")
plt.axis('off') # Hide axes for cleaner display
plt.show()
```



```
import tensorflow as tf
from tensorflow.keras import layers, Model
from tensorflow.keras.applications import ResNet50
```

```
base_model.trainable = False    # freezing backbone
```

```
print("Classes:", train_gen.class_indices) # Check class labels
```

```
num_classes=len(train_gen.class_indices) # Number of classes in the train dataset
num_classes
```

- Adding an appropriate number of dense layers with an activation function
- Using dropout for regularization

```
x = base_model.output
x = layers.GlobalAveragePooling2D()(x)

x = layers.Dense(256, activation='relu')(x)
x = layers.Dropout(0.5)(x)

outputs = layers.Dense(num_classes, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=outputs)
```

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Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 128, 128, 3)	0	–
conv1_pad (ZeroPadding2D)	(None, 134, 134, 3)	0	input_layer_1[0]...
conv1_conv (Conv2D)	(None, 64, 64, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio...	(None, 64, 64, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 64, 64, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 66, 66, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 32, 32, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 32, 32, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_1_c...
conv2_block1_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_1_b...
conv2_block1_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block1_1_r...
conv2_block1_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_2_c...
conv2_block1_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_2_b...
conv2_block1_0_conv (Conv2D)	(None, 32, 32, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block1_2_r...
conv2_block1_0_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_0_c...
conv2_block1_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_3_c...
conv2_block1_add (Add)	(None, 32, 32, 256)	0	conv2_block1_0_b... conv2_block1_3_b...
conv2_block1_out (Activation)	(None, 32, 32, 256)	0	conv2_block1_add...
conv2_block2_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block1_out...
conv2_block2_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_1_c...
conv2_block2_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_1_b...
conv2_block2_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block2_1_r...
conv2_block2_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_2_c...
conv2_block2_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_2_b...
conv2_block2_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block2_2_r...
conv2_block2_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block2_3_c...
conv2_block2_add (Add)	(None, 32, 32, 256)	0	conv2_block1_out... conv2_block2_3_b...
conv2_block2_out (Activation)	(None, 32, 32, 256)	0	conv2_block2_add...
conv2_block3_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block2_out...
conv2_block3_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_1_c...
conv2_block3_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_1_b...

(Activation)	64)		
conv2_block3_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block3_1_r...
conv2_block3_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_2_c...
conv2_block3_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_2_b...
conv2_block3_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block3_2_r...
conv2_block3_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block3_3_c...
conv2_block3_add (Add)	(None, 32, 32, 256)	0	conv2_block2_out... conv2_block3_3_b...
conv2_block3_out (Activation)	(None, 32, 32, 256)	0	conv2_block3_add...
conv3_block1_1_conv (Conv2D)	(None, 16, 16, 128)	32,896	conv2_block3_out...
conv3_block1_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_1_c...
conv3_block1_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_1_b...
conv3_block1_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block1_1_r...
conv3_block1_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_2_c...
conv3_block1_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_2_b...
conv3_block1_0_conv (Conv2D)	(None, 16, 16, 512)	131,584	conv2_block3_out...
conv3_block1_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block1_2_r...
conv3_block1_0_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_0_c...
conv3_block1_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_3_c...
conv3_block1_add (Add)	(None, 16, 16, 512)	0	conv3_block1_0_b... conv3_block1_3_b...
conv3_block1_out (Activation)	(None, 16, 16, 512)	0	conv3_block1_add...
conv3_block2_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block1_out...
conv3_block2_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_1_c...
conv3_block2_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_1_b...
conv3_block2_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block2_1_r...
conv3_block2_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_2_c...
conv3_block2_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_2_b...
conv3_block2_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block2_2_r...
conv3_block2_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block2_3_c...
conv3_block2_add (Add)	(None, 16, 16, 512)	0	conv3_block1_out... conv3_block2_3_b...
conv3_block2_out (Activation)	(None, 16, 16, 512)	0	conv3_block2_add...
conv3_block3_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block2_out...
conv3_block3_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_1_c...
conv3_block3_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_1_b...

conv3_block3_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block3_1_r...
conv3_block3_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_2_c...
conv3_block3_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_2_b...
conv3_block3_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block3_2_r...
conv3_block3_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block3_3_c...
conv3_block3_add (Add)	(None, 16, 16, 512)	0	conv3_block2_out... conv3_block3_3_b...
conv3_block3_out (Activation)	(None, 16, 16, 512)	0	conv3_block3_add...
conv3_block4_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block3_out...
conv3_block4_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_1_c...
conv3_block4_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_1_b...
conv3_block4_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block4_1_r...
conv3_block4_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_2_c...
conv3_block4_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_2_b...
conv3_block4_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block4_2_r...
conv3_block4_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block4_3_c...
conv3_block4_add (Add)	(None, 16, 16, 512)	0	conv3_block3_out... conv3_block4_3_b...
conv3_block4_out (Activation)	(None, 16, 16, 512)	0	conv3_block4_add...
conv4_block1_1_conv (Conv2D)	(None, 8, 8, 256)	131,328	conv3_block4_out...
conv4_block1_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_1_c...
conv4_block1_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_1_b...
conv4_block1_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block1_1_r...
conv4_block1_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_2_c...
conv4_block1_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_2_b...
conv4_block1_0_conv (Conv2D)	(None, 8, 8, 1024)	525,312	conv3_block4_out...
conv4_block1_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block1_2_r...
conv4_block1_0_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_0_c...
conv4_block1_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_3_c...
conv4_block1_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_0_b... conv4_block1_3_b...
conv4_block1_out (Activation)	(None, 8, 8, 1024)	0	conv4_block1_add...
conv4_block2_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block1_out...
conv4_block2_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block2_1_c...
conv4_block2_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_1_b...
conv4_block2_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block2_1_r...

conv4_block2_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block2_2_c...
conv4_block2_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_2_b...
conv4_block2_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block2_2_r...
conv4_block2_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block2_3_c...
conv4_block2_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_out... conv4_block2_3_b...
conv4_block2_out (Activation)	(None, 8, 8, 1024)	0	conv4_block2_add...
conv4_block3_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block2_out...
conv4_block3_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block3_1_c...
conv4_block3_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_1_b...
conv4_block3_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block3_1_r...
conv4_block3_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block3_2_c...
conv4_block3_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_2_b...
conv4_block3_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block3_2_r...
conv4_block3_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block3_3_c...
conv4_block3_add (Add)	(None, 8, 8, 1024)	0	conv4_block2_out... conv4_block3_3_b...
conv4_block3_out (Activation)	(None, 8, 8, 1024)	0	conv4_block3_add...
conv4_block4_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block3_out...
conv4_block4_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block4_1_c...
conv4_block4_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_1_b...
conv4_block4_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block4_1_r...
conv4_block4_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block4_2_c...
conv4_block4_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_2_b...
conv4_block4_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block4_2_r...
conv4_block4_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block4_3_c...
conv4_block4_add (Add)	(None, 8, 8, 1024)	0	conv4_block3_out... conv4_block4_3_b...
conv4_block4_out (Activation)	(None, 8, 8, 1024)	0	conv4_block4_add...
conv4_block5_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block4_out...
conv4_block5_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block5_1_c...
conv4_block5_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_1_b...
conv4_block5_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block5_1_r...
conv4_block5_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block5_2_c...
conv4_block5_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_2_b...
conv4_block5_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block5_2_r...

(Conv2D)	(None, 8, 8, 1024)	4,096	conv4_block5_3_c...
conv4_block5_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	0	conv4_block4_out...
conv4_block5_add (Add)	(None, 8, 8, 1024)	0	conv4_block5_add...
conv4_block5_out (Activation)	(None, 8, 8, 1024)	262,400	conv4_block5_out...
conv4_block6_1_conv (Conv2D)	(None, 8, 8, 256)	1,024	conv4_block6_1_c...
conv4_block6_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	0	conv4_block6_1_b...
conv4_block6_1_relu (Activation)	(None, 8, 8, 256)	590,080	conv4_block6_1_r...
conv4_block6_2_conv (Conv2D)	(None, 8, 8, 256)	1,024	conv4_block6_2_c...
conv4_block6_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	0	conv4_block6_2_b...
conv4_block6_2_relu (Activation)	(None, 8, 8, 256)	263,168	conv4_block6_2_r...
conv4_block6_3_conv (Conv2D)	(None, 8, 8, 1024)	4,096	conv4_block6_3_c...
conv4_block6_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	0	conv4_block5_out...
conv4_block6_add (Add)	(None, 8, 8, 1024)	0	conv4_block6_add...
conv4_block6_out (Activation)	(None, 8, 8, 1024)	524,800	conv4_block6_out...
conv5_block1_1_conv (Conv2D)	(None, 4, 4, 512)	2,048	conv5_block1_1_c...
conv5_block1_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	0	conv5_block1_1_b...
conv5_block1_1_relu (Activation)	(None, 4, 4, 512)	2,359,808	conv5_block1_1_r...
conv5_block1_2_conv (Conv2D)	(None, 4, 4, 512)	2,048	conv5_block1_2_c...
conv5_block1_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	0	conv5_block1_2_b...
conv5_block1_2_relu (Activation)	(None, 4, 4, 512)	2,099,200	conv4_block6_out...
conv5_block1_0_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block1_2_r...
conv5_block1_3_conv (Conv2D)	(None, 4, 4, 2048)	8,192	conv5_block1_0_c...
conv5_block1_0_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block1_3_c...
conv5_block1_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)	0	conv5_block1_0_b...
conv5_block1_add (Add)	(None, 4, 4, 2048)	0	conv5_block1_3_b...
conv5_block1_out (Activation)	(None, 4, 4, 2048)	1,049,088	conv5_block1_add...
conv5_block2_1_conv (Conv2D)	(None, 4, 4, 512)	2,048	conv5_block1_out...
conv5_block2_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	0	conv5_block2_1_c...
conv5_block2_1_relu (Activation)	(None, 4, 4, 512)	2,359,808	conv5_block2_1_b...
conv5_block2_2_conv (Conv2D)	(None, 4, 4, 512)	2,048	conv5_block2_1_r...
conv5_block2_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	0	conv5_block2_2_c...
conv5_block2_2_relu (Activation)	(None, 4, 4, 512)	1,050,624	conv5_block2_2_b...
conv5_block2_3_conv (Conv2D)	(None, 4, 4, 2048)	8,192	conv5_block2_2_r...
conv5_block2_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)		conv5_block2_3_c...

5. Compilation of the model	(Batch Normalization)	2048		
conv5_block2_add	(None, 4, 4, 4)	0	conv5_block1_out...	

```
# model compilation
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

```
# fitting the data with the model
history = model.fit(
    train_gen,
    validation_data=val_gen,
    batch_size=64,
    epochs=10
)
```

conv5_block3_2_relu	(None, 4, 4, 512)	0	conv5_block3_2_b...	
Epoch 1/10	(Activation)	2512s 10s/step	accuracy: 0.2642	loss: 2.1002 - val_accuracy: 0.3741 - val_loss:
Epoch 2/10	conv5_block3_3_conv	(None, 4, 4, 2048)	1,050,624	conv5_block3_2_r...
Epoch 3/10	conv5_block3_3_bn	(None, 4, 4, 2048)	631s 2s/step	accuracy: 0.3422 - loss: 1.8909 - val_accuracy: 0.3976 - val_loss: 1.
Epoch 4/10	conv5_block3_3_bn	(None, 4, 4, 2048)	679s 2s/step	accuracy: 0.3785 - loss: 1.7250 - val_accuracy: 0.4060 - val_loss: 1.
Epoch 5/10	conv5_block3_add	(None, 4, 4, 2048)	625s 2s/step	accuracy: 0.4035 - loss: 1.7250 - val_accuracy: 0.4182 - val_loss: 1.
Epoch 6/10	conv5_block3_out	(None, 4, 4, 2048)	641s 2s/step	accuracy: 0.4393 - loss: 1.6217 - val_accuracy: 0.4814 - val_loss: 1.
Epoch 7/10	global_average_pooling2d	(None, 2048)	644s 2s/step	accuracy: 0.4669 - loss: 1.5771 - val_accuracy: 0.4907 - val_loss: 1.
Epoch 8/10	global_average_pooling2d	(None, 2048)	627s 2s/step	accuracy: 0.4625 - loss: 1.5777 - val_accuracy: 0.5039 - val_loss: 1.
Epoch 9/10	dense_0 (Dense)	(None, 256)	524,544	global_average_p...
Epoch 10/10	dense_1 (Dropout)	(None, 256)	0	dense_1[0][0]
dense 2 (Dense)	(None, 10)	638s 2s/step	accuracy: 0.4903 - loss: 1.5085 - val_accuracy: 0.5362 - val_loss: 1.	dropout 1[0][0]

Total params: 24,114,826 (91.99 MB)
 Model accuracy improves with the epochs and validation loss decreases with the epochs. Therefore, more epochs can fix into the better performance.
 Non-trainable params: 23,587,712 (89.98 MB)

```
# checking the loss and accuracy with the test data
loss, accuracy = model.evaluate(test_gen)
print("Test Loss:", loss)
print("Test Accuracy:", accuracy)
```

47/47 ————— 98s 2s/step - accuracy: 0.4454 - loss: 1.6280
 Test Loss: 1.6280131340026855
 Test Accuracy: 0.445423722671509

```
# Prediction of the class from the test data
y_pred = model.predict(test_gen)
```

47/47 ————— 96s 2s/step

```
# prediction of the class from the test data set the from the predicted probability of y_pred values
import numpy as np
y_pred_classes = np.argmax(y_pred, axis=1)
y_pred_classes
```

array([3, 0, 0, ..., 0, 9, 7])

```
# True classes from teh test data
y_true = test_gen.classes
y_true
```

array([0, 0, 0, ..., 9, 9, 9], dtype=int32)

```
# Import an random images from the test director and check the predicion with the true values
import random

file_paths = test_gen.filepaths # Map the filepath to its original class index using test_gen's structure
img_path = random.choice(file_paths)

# load image using cv2.imread
img = cv2.imread(img_path)
```



```

# Check if image was loaded successfully
if img is None:
    print(f"Error: Unable to load image at {img_path}")
else:
    # Convert BGR to RGB ( As OpenCV loads images in BGR by default)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    # Resize image to the target size (128, 128) used during training
    target_size = (128, 128) # As defined in img_size earlier
    img_resized = cv2.resize(img, target_size)

    # Convert image to array and normalize
    img_array = np.array(img_resized) / 255.0
    img_array = np.expand_dims(img_array, axis=0)

    # prediction of classes
    prediction = model.predict(img_array)
    pred_class = np.argmax(prediction, axis=1)[0]

    # labels for classes
    class_labels = dict((v, k) for k, v in test_gen.class_indices.items())

    predicted_label = class_labels[pred_class]

    # Find the true label corresponding to the image path
    # For simplicity, assuming file_paths.index(img_path) gives the correct index into test_gen.classes
    true_label_index = test_gen.filepaths.index(img_path)
    true_label = class_labels[test_gen.classes[true_label_index]]

    print("Predicted:", predicted_label)
    print("Actual:", true_label)

    # Display the image with its predicted and actual labels
    plt.imshow(img_resized)
    plt.title(f"Predicted: {predicted_label}\nActual: {true_label}")
    plt.axis('off')

    plt.show()

```

1/1 ————— 0s 134ms/step
 Predicted: dome(inner)
 Actual: dome(inner)

Predicted: dome(inner)
 Actual: dome(inner)



For some images it correctly predict the class but for other images it can not predict the right class. It is possibly due to the poor (~50%) training and validation accuracy. The testing accuracy is also poor, about 44%. We need to check with the augmented data.

Model performance with Augmented image data

```

# Augmentation of the images by changing the zoom and brightness modulation
advanced_datagen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2,
    zoom_range=0.05,

```

```
brightness_range=[0.9, 1.1]
)
```

```
# Dataset assignment with passing the label of augmentation
train_gen_aug = advanced_datagen.flow_from_directory(
    train_dir,
    target_size=img_size,
    batch_size=batch_size,
    class_mode='categorical',
    subset='training'
)
```

```
val_gen_aug = advanced_datagen.flow_from_directory(
    train_dir,
    target_size=img_size,
    batch_size=batch_size,
    class_mode='categorical',
    subset='validation'
)
```

```
# We don't need any augmentation for the tes data set
```

```
Found 8193 images belonging to 10 classes.
Found 2042 images belonging to 10 classes.
```

```
# Function to check the number of images in each class subfolder for train and test data set
import os
```

```
def count_images(folder):
    class_counts = {}

    for class_name in os.listdir(folder):
        class_path = os.path.join(folder, class_name)

        if os.path.isdir(class_path):
            num_images = len(os.listdir(class_path))
            class_counts[class_name] = num_images

    return class_counts
```

```
train_counts = count_images(train_dir)
test_counts = count_images(test_dir)
```

```
print("Train:", train_counts)
print("Test:", test_counts)
```

```
# Count of the number of images in each train class subfolder
class_labels = {v: k for k, v in train_gen.class_indices.items()}

train_distribution = {class_labels[k]: v for k, v in train_counts.items()}

print("Train distribution (label-wise):")
for k, v in train_distribution.items():
    print(f"{k}: {v}")
```

```
Train distribution (label-wise):
altar: 664
apse: 412
bell_tower: 848
column: 1536
dome(inner): 493
dome(outer): 942
flying_buttress: 326
gargoyle: 1257
stained_glass: 827
vault: 888
```

```
# Count of the number of images in each train class subfolder
class_labels = {v: k for k, v in test_gen.class_indices.items()}

test_distribution = {class_labels[k]: v for k, v in test_counts.items()}

print("Train distribution (label-wise):")
for k, v in test_distribution.items():
    print(f"{k}: {v}")
```

```
Train distribution (label-wise):
altar: 140
apse: 57
bell_tower: 170
column: 210
dome(inner): 86
dome(outer): 168
```

```
flying_buttress: 78  
gargoyle: 240  
stained_glass: 162  
vault: 164
```

Using the same model for Training

```
# same base model for transfer learning  
base_model = ResNet50(  
    weights='imagenet',    # pretrained weights used  
    include_top=False,    # removal of classification head  
    input_shape=(128, 128, 3)  
)
```

```
base_model.trainable = False    # freeze backbone
```

```
# same base model output for pass into the Dense layer  
x = base_model.output  
x = layers.GlobalAveragePooling2D()(x)
```

```
x = layers.Dense(256, activation='relu')(x)  
x = layers.Dropout(0.5)(x)
```

```
outputs = layers.Dense(num_classes, activation='softmax')(x)
```

```
model_res50 = Model(inputs=base_model.input, outputs=outputs) # Model renamed for better handling
```

```
model_res50.summary()
```


Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 128, 128, 3)	0	–
conv1_pad (ZeroPadding2D)	(None, 134, 134, 3)	0	input_layer_1[0]...
conv1_conv (Conv2D)	(None, 64, 64, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio...	(None, 64, 64, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 64, 64, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 66, 66, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 32, 32, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 32, 32, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_1_c...
conv2_block1_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_1_b...
conv2_block1_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block1_1_r...
conv2_block1_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block1_2_c...
conv2_block1_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block1_2_b...
conv2_block1_0_conv (Conv2D)	(None, 32, 32, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block1_2_r...
conv2_block1_0_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_0_c...
conv2_block1_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block1_3_c...
conv2_block1_add (Add)	(None, 32, 32, 256)	0	conv2_block1_0_b... conv2_block1_3_b...
conv2_block1_out (Activation)	(None, 32, 32, 256)	0	conv2_block1_add...
conv2_block2_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block1_out...
conv2_block2_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_1_c...
conv2_block2_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_1_b...
conv2_block2_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block2_1_r...
conv2_block2_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block2_2_c...
conv2_block2_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block2_2_b...
conv2_block2_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block2_2_r...
conv2_block2_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block2_3_c...
conv2_block2_add (Add)	(None, 32, 32, 256)	0	conv2_block1_out... conv2_block2_3_b...
conv2_block2_out (Activation)	(None, 32, 32, 256)	0	conv2_block2_add...
conv2_block3_1_conv (Conv2D)	(None, 32, 32, 64)	16,448	conv2_block2_out...
conv2_block3_1_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_1_c...
conv2_block3_1_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_1_b...

(Activation)	64)		
conv2_block3_2_conv (Conv2D)	(None, 32, 32, 64)	36,928	conv2_block3_1_r...
conv2_block3_2_bn (BatchNormalizatio...	(None, 32, 32, 64)	256	conv2_block3_2_c...
conv2_block3_2_relu (Activation)	(None, 32, 32, 64)	0	conv2_block3_2_b...
conv2_block3_3_conv (Conv2D)	(None, 32, 32, 256)	16,640	conv2_block3_2_r...
conv2_block3_3_bn (BatchNormalizatio...	(None, 32, 32, 256)	1,024	conv2_block3_3_c...
conv2_block3_add (Add)	(None, 32, 32, 256)	0	conv2_block2_out... conv2_block3_3_b...
conv2_block3_out (Activation)	(None, 32, 32, 256)	0	conv2_block3_add...
conv3_block1_1_conv (Conv2D)	(None, 16, 16, 128)	32,896	conv2_block3_out...
conv3_block1_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_1_c...
conv3_block1_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_1_b...
conv3_block1_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block1_1_r...
conv3_block1_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block1_2_c...
conv3_block1_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block1_2_b...
conv3_block1_0_conv (Conv2D)	(None, 16, 16, 512)	131,584	conv2_block3_out...
conv3_block1_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block1_2_r...
conv3_block1_0_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_0_c...
conv3_block1_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block1_3_c...
conv3_block1_add (Add)	(None, 16, 16, 512)	0	conv3_block1_0_b... conv3_block1_3_b...
conv3_block1_out (Activation)	(None, 16, 16, 512)	0	conv3_block1_add...
conv3_block2_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block1_out...
conv3_block2_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_1_c...
conv3_block2_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_1_b...
conv3_block2_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block2_1_r...
conv3_block2_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block2_2_c...
conv3_block2_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block2_2_b...
conv3_block2_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block2_2_r...
conv3_block2_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block2_3_c...
conv3_block2_add (Add)	(None, 16, 16, 512)	0	conv3_block1_out... conv3_block2_3_b...
conv3_block2_out (Activation)	(None, 16, 16, 512)	0	conv3_block2_add...
conv3_block3_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block2_out...
conv3_block3_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_1_c...
conv3_block3_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_1_b...

conv3_block3_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block3_1_r...
conv3_block3_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block3_2_c...
conv3_block3_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block3_2_b...
conv3_block3_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block3_2_r...
conv3_block3_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block3_3_c...
conv3_block3_add (Add)	(None, 16, 16, 512)	0	conv3_block2_out... conv3_block3_3_b...
conv3_block3_out (Activation)	(None, 16, 16, 512)	0	conv3_block3_add...
conv3_block4_1_conv (Conv2D)	(None, 16, 16, 128)	65,664	conv3_block3_out...
conv3_block4_1_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_1_c...
conv3_block4_1_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_1_b...
conv3_block4_2_conv (Conv2D)	(None, 16, 16, 128)	147,584	conv3_block4_1_r...
conv3_block4_2_bn (BatchNormalizatio...	(None, 16, 16, 128)	512	conv3_block4_2_c...
conv3_block4_2_relu (Activation)	(None, 16, 16, 128)	0	conv3_block4_2_b...
conv3_block4_3_conv (Conv2D)	(None, 16, 16, 512)	66,048	conv3_block4_2_r...
conv3_block4_3_bn (BatchNormalizatio...	(None, 16, 16, 512)	2,048	conv3_block4_3_c...
conv3_block4_add (Add)	(None, 16, 16, 512)	0	conv3_block3_out... conv3_block4_3_b...
conv3_block4_out (Activation)	(None, 16, 16, 512)	0	conv3_block4_add...
conv4_block1_1_conv (Conv2D)	(None, 8, 8, 256)	131,328	conv3_block4_out...
conv4_block1_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_1_c...
conv4_block1_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_1_b...
conv4_block1_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block1_1_r...
conv4_block1_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block1_2_c...
conv4_block1_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block1_2_b...
conv4_block1_0_conv (Conv2D)	(None, 8, 8, 1024)	525,312	conv3_block4_out...
conv4_block1_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block1_2_r...
conv4_block1_0_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_0_c...
conv4_block1_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block1_3_c...
conv4_block1_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_0_b... conv4_block1_3_b...
conv4_block1_out (Activation)	(None, 8, 8, 1024)	0	conv4_block1_add...
conv4_block2_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block1_out...
conv4_block2_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block2_1_c...
conv4_block2_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_1_b...
conv4_block2_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block2_1_r...

conv4_block2_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block2_2_c...
conv4_block2_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block2_2_b...
conv4_block2_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block2_2_r...
conv4_block2_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block2_3_c...
conv4_block2_add (Add)	(None, 8, 8, 1024)	0	conv4_block1_out... conv4_block2_3_b...
conv4_block2_out (Activation)	(None, 8, 8, 1024)	0	conv4_block2_add...
conv4_block3_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block2_out...
conv4_block3_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block3_1_c...
conv4_block3_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_1_b...
conv4_block3_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block3_1_r...
conv4_block3_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block3_2_c...
conv4_block3_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block3_2_b...
conv4_block3_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block3_2_r...
conv4_block3_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block3_3_c...
conv4_block3_add (Add)	(None, 8, 8, 1024)	0	conv4_block2_out... conv4_block3_3_b...
conv4_block3_out (Activation)	(None, 8, 8, 1024)	0	conv4_block3_add...
conv4_block4_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block3_out...
conv4_block4_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block4_1_c...
conv4_block4_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_1_b...
conv4_block4_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block4_1_r...
conv4_block4_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block4_2_c...
conv4_block4_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block4_2_b...
conv4_block4_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block4_2_r...
conv4_block4_3_bn (BatchNormalization)	(None, 8, 8, 1024)	4,096	conv4_block4_3_c...
conv4_block4_add (Add)	(None, 8, 8, 1024)	0	conv4_block3_out... conv4_block4_3_b...
conv4_block4_out (Activation)	(None, 8, 8, 1024)	0	conv4_block4_add...
conv4_block5_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block4_out...
conv4_block5_1_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block5_1_c...
conv4_block5_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_1_b...
conv4_block5_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block5_1_r...
conv4_block5_2_bn (BatchNormalization)	(None, 8, 8, 256)	1,024	conv4_block5_2_c...
conv4_block5_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block5_2_b...
conv4_block5_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block5_2_r...

(Conv2D)	(None, 8, 8, 1024)	4,096	conv4_block5_3_c...
conv4_block5_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	0	conv4_block4_out...
conv4_block5_add (Add)	(None, 8, 8, 1024)	0	conv4_block5_3_b...
conv4_block5_out (Activation)	(None, 8, 8, 1024)	0	conv4_block5_add...
conv4_block6_1_conv (Conv2D)	(None, 8, 8, 256)	262,400	conv4_block5_out...
conv4_block6_1_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block6_1_c...
conv4_block6_1_relu (Activation)	(None, 8, 8, 256)	0	conv4_block6_1_b...
conv4_block6_2_conv (Conv2D)	(None, 8, 8, 256)	590,080	conv4_block6_1_r...
conv4_block6_2_bn (BatchNormalizatio...	(None, 8, 8, 256)	1,024	conv4_block6_2_c...
conv4_block6_2_relu (Activation)	(None, 8, 8, 256)	0	conv4_block6_2_b...
conv4_block6_3_conv (Conv2D)	(None, 8, 8, 1024)	263,168	conv4_block6_2_r...
conv4_block6_3_bn (BatchNormalizatio...	(None, 8, 8, 1024)	4,096	conv4_block6_3_c...
conv4_block6_add (Add)	(None, 8, 8, 1024)	0	conv4_block5_out...
conv4_block6_out (Activation)	(None, 8, 8, 1024)	0	conv4_block6_add...
conv5_block1_1_conv (Conv2D)	(None, 4, 4, 512)	524,800	conv4_block6_out...
conv5_block1_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block1_1_c...
conv5_block1_1_relu (Activation)	(None, 4, 4, 512)	0	conv5_block1_1_b...
conv5_block1_2_conv (Conv2D)	(None, 4, 4, 512)	2,359,808	conv5_block1_1_r...
conv5_block1_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block1_2_c...
conv5_block1_2_relu (Activation)	(None, 4, 4, 512)	0	conv5_block1_2_b...
conv5_block1_0_conv (Conv2D)	(None, 4, 4, 2048)	2,099,200	conv4_block6_out...
conv5_block1_3_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block1_2_r...
conv5_block1_0_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block1_0_c...
conv5_block1_3_bn (BatchNormalizatio...	(None, 4, 4, 2048)	8,192	conv5_block1_3_c...
conv5_block1_add (Add)	(None, 4, 4, 2048)	0	conv5_block1_0_b...
conv5_block1_out (Activation)	(None, 4, 4, 2048)	0	conv5_block1_3_b...
conv5_block2_1_conv (Conv2D)	(None, 4, 4, 512)	1,049,088	conv5_block1_add...
conv5_block2_1_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block1_out...
conv5_block2_1_relu (Activation)	(None, 4, 4, 512)	0	conv5_block2_1_c...
conv5_block2_2_conv (Conv2D)	(None, 4, 4, 512)	2,359,808	conv5_block2_1_b...
conv5_block2_2_bn (BatchNormalizatio...	(None, 4, 4, 512)	2,048	conv5_block2_1_r...
conv5_block2_2_relu (Activation)	(None, 4, 4, 512)	0	conv5_block2_2_c...
conv5_block2_3_conv (Conv2D)	(None, 4, 4, 2048)	1,050,624	conv5_block2_2_b...
conv5_block2_3_bn	(None, 4, 4, 2048)	8,192	conv5_block2_2_r...
conv5_block2_3_c...	(None, 4, 4, 2048)	8,192	conv5_block2_3_c...

conv5 block3 1 conv	(None, 4, 4, 512)	1,049,088	conv5 block2 out...
---------------------	-------------------	-----------	---------------------

```
callbacks = [
    EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True),
    ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=2, verbose=1),
    ModelCheckpoint("best_resnet50.keras", monitor='val_accuracy', save_best_only=True, verbose=1)
]
```

| BatchNormalization

Epoch 10: finished saving model to best_resnet50.keras
257/257  793s 3s/step - accuracy: 0.4715 - loss: 1.5434 - val accuracy: 0.5162 - val_loss: 1.5434

```
# Check the loss and accuracy with the test data
loss, accuracy = model_res50.evaluate(test_gen)
print("Test Loss:", loss)
print("Test Accuracy:", accuracy)
```

```
47/47 ————— 104s 2s/step - accuracy: 0.4542 - loss: 1.5939
Test Loss: 1.5939288139343262
Test Accuracy: 0.45423728227615356
```

```
# Check the model performance with epochs

import matplotlib.pyplot as plt

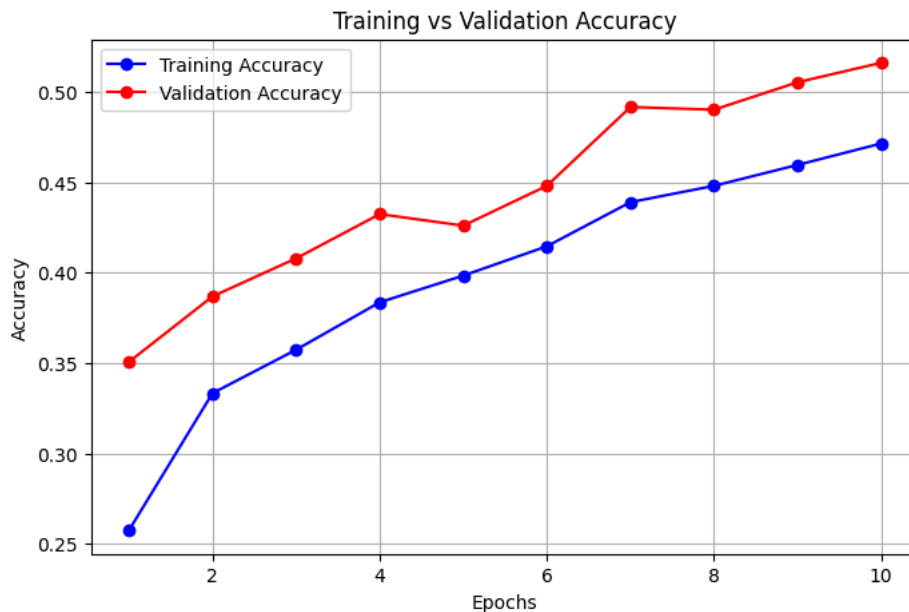
# Extract values
train_acc = history_res50.history['accuracy']
val_acc = history_res50.history['val_accuracy']

epochs = range(1, len(train_acc) + 1)

plt.figure(figsize=(8,5))
plt.plot(epochs, train_acc, 'bo-', label='Training Accuracy')
plt.plot(epochs, val_acc, 'ro-', label='Validation Accuracy')

plt.title('Training vs Validation Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)

plt.show()
```



Training and validation accuracy do not improve so much with 10 epochs but saturation point not achieved with 10 epochs. Possibly due to the bas model training performance.

```
train_loss = history_res50.history['loss']
val_loss = history_res50.history['val_loss']

plt.figure(figsize=(8,5))
plt.plot(epochs, train_loss, 'bo-', label='Training Loss')
plt.plot(epochs, val_loss, 'ro-', label='Validation Loss')

plt.title('Training vs Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.show()
```